MILITARY SPECIFICATION

CASE, CARTRIDGE, 90MM, XM200

1. SCOPE

1.1 This specification covers Case, Cartridge, 90MM, XM200.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification For
- MIL-P-10025 - Packing, Packing and Marking for Interplant Shipment of Inert Ammunition Components: General Specification For

STANDARDS
Federal
- Federal Test Method Standard No. 151 - Metals; Test Methods

Military
- MIL-STD- 105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD- 109 - Quality Assurance Terms and Definitions
- MIL-STD- 639 - Surface Roughness Values Preferred for Army Ordnance Materiel
- MIL-STD-1167 - Ammunition Data Cards
- MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

DRAWINGS
Munitions Command
F10535933 - Case, Cartridge, 90MM, XM200
F-7548235 - Cartons, Ammunition for Interplant Shipment of Steel Cartridge Cases, 37MM to 120MM, Inclusive
IEL10535933 - Index to Inspection Equipment List for Case, Cartridge, 90MM, XM200

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with all the requirements of the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for submission of first article sample by the contractor shall be as specified in the contract.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Mechanical properties.- The mechanical properties specified on the applicable drawing and referenced specifications shall be uniform throughout each section.

3.4.1 Thermal treatment.- The entire cartridge case shall be thermally treated after the heading operation by heating above the upper critical temperature point, quenching and tempering to the desired properties.

3.4.1.1 Stress relief.- The cartridge case shall be adequately stress relieved at the times and temperatures suitable for development of desired properties.

3.5 Protective coating.- Protective coating as specified on the applicable drawings and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration.

3.6 Maximum profile.- The maximum profile of the cartridge case shall comply with the requirements of the applicable drawing.

3.7 Functional.- The cartridge case shall be fired in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (For information only)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.7.1 Obturation.- The cartridge case shall effectively seal the breech of the weapon against gases generated during firing.

3.7.2 Soundness.- After firing, the cartridge case shall be free of cracks or splits, except as permitted by Table II.

3.7.3 Ejection.- After firing, the cartridge case shall be ejected by the service extractors when the weapon is operated in the normal manner. (See 6.3)

3.7.4 Primer retention.- The cartridge case shall retain its primer during firing and ejection.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.8.1.1 Flow of metal.- The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.8.2 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.8.3 Foreign matter.- No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 274 cartridge cases as follows:

254 cases coated
20 cases uncoated

250 of the above coated cases shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Four power (4X) magnification shall be used in performing visual examination for cracks as specified. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.1 Case, cartridge, Drawing F-10535933 - Prior to Coating.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of pitch diameter of thread with body	Gage
102.	Diameter of flange, min.	Gage
103.	Diameter of shoulder under flange, min.	Gage
104.	Diameter of mouth	Gage
105.	Thickness of flange, min.	Gage
106.	Thickness of flange and shoulder, min.	Gage
107.	Thickness of head, min.	Gage
108.	Thickness of wall at basic location near head	Gage
109.	Thickness of wall at basic location near mid body	Gage
110.	Thickness of wall at basic location near mouth	Gage
111.	Thickness of wall at mouth	Gage
112.	Contour of inside of head and sidewall, not smooth	Visual
113.	Metal defective (unless otherwise classified)	
MINOR		
201.	Diameter of body under shoulder, min.	Gage
202.	Total length, min.	Gage
203.	Location of first basic diameter from head end, min.	Gage
204.	Location of second basic diameter, from head end, min.	Gage
205.	Location of third basic diameter from head end, min.	Gage
206.	Location of fourth basic diameter from head end, min.	Gage
207.	Radii or chamfers, missing or incorrect	Visual (See 6.1)
208.	Surface finish, improper	Visual (See 6.1)
209.	Burr or sharp corner, except where permitted	Visual
210.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.2 Case, cartridge, drawing F-10535933 - After coating.

Defect	Method of Inspection
CRITICAL	
1. Cracks in sidewall within 8 inches of head - 4X Magnification	
MAJOR	
101. Concentricity of primer hole counterbore with pitch diameter of thread	Gage
102. Pitch diameter of primer hold thread, max.	Gage
103. Minor diameter of primer hole thread, max.	Gage
104. Depth of primer hole counterbore	Gage
105. Total length, max.	Gage
106. Protective coating, inadequate or damaged	Visual
107. Metal defective (unless otherwise classified)	Visual
MINOR	
201. Pitch diameter of primer hole thread, min.	Gage
202. Diameter of primer hole counterbore	Gage
203. Dent or damage	Visual
204. Thread damaged or not full	Visual
205. Varnish in thread or primer hole counterbore	Visual
206. Silicone compound missing from thread or primer hole counterbore	Visual
207. Marking missing, incorrect or illegible	Visual

4.3.2.3 Carton, packing, Drawing F-7548235 - Prior to Sealing.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Incorrect number in carton	Visual
203. Separator, missing or damaged	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.4 Carton, packing, Drawing F-7548235 - After sealing.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton improperly sealed	Visual
203. Contents loose	Manual and Visual
204. Marking, missing, incorrect or illegible	Visual

4.3.3 Maximum profile gage (A major defect).- Each cartridge case shall be gaged in a Government certified maximum profile gage. Failure of the cartridge case to comply with the requirements of 3.6 shall be cause for rejection.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.4.1 Mechanical properties.

4.3.4.1.1 Hardness.- Twenty cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of 3.4 shall be cause for rejection of the sample or lot.

4.3.4.1.1.1 Thermal treatment.- Each cartridge case shall be subjected to a hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.4.1 or indications that the cartridge case failed to harden shall be cause for rejection of the cartridge case.

4.3.4.1.2 Yield strength, tensile strength and elongation.- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. The results shall be recorded for information only. The cartridge cases used for the hardness test may be used for this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4.1.3 Macro examination.- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be examined for folds and cold shuts. Failure of one or more cartridge cases to comply with the requirements of 3.8.1.1 shall be cause for rejection of the sample or lot. The cartridge cases used for the hardness test may be used for this test.

4.3.4.1.4 Protective coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.4.2 Functional.- The following functional tests will be performed by the Government.

4.3.4.2.1 First article sample.

4.3.4.2.1.1 Initial production sample.- The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table II, except no retest will be permitted.

Table I

Test Phase	Quantity to be Tested
Excess pressure phase	100
Cold temperature phase	50
Excess, excess pressure phase (for information only)	100

4.3.4.2.2 Production lot sample.- One hundred cartridge cases from each production lot of 25,000 or less, will be ballistically tested in the excess pressure phase. When lots exceed 25,000 cartridge cases, four additional cartridge cases will be tested for every 1,000 or fraction thereof, above 25,000. Acceptance and rejection criteria shall be in accordance with Table II. The lot shall be considered eligible for retest, using double the number of samples, provided that after firing no cartridge case contains crack(s) or split(s) within one caliber of the head, and the rejection criteria of Table II has not been attained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

Table II

Defect Characteristic	First Sample Accept	First Sample Reject	Cumulative Sample Accept	Cumulative Sample Reject
1. Obturation	0	2	1	2
2. Splits or cracks within one caliber of the head	0	1	Not permitted	
3. Splits or cracks in sidewall more than one caliber from the head	1	3	2	3
4. Ejection	0	2	1	2
5. Primer retention	0	2	1	2

4.3.5 Inspection equipment.- Index of Inspection Equipment Lists No. 10535933 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Hardness.- This test shall be performed on locations specified by the applicable drawing in accordance with the procedures prescribed for Rockwell hardness testing by Federal Test Method Standard No. 151.

4.4.1.1 Thermal treatment.- This test shall be performed on a comparator of a nondestructive type as listed on the referenced inspection equipment list.

4.4.2 Yield strength, tensile strength and elongation.- The test speciments shall be taken from the cartridge case in accordance with the standard positions and size of tensile specimen chart, Figure I. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed for these tests by Federal Test Method Standard No. 151.

4.4.3 Macro examination.- The cartridge case shall be cut in a 90 degree section along the axis through the primer hole. The specimen shall be etched in accordance with the procedures prescribed for this test by Federal Test Method Standard No. 151.

4.4.4 Protective coating.- The procedures and equipment for performing these tests shall be as prescribed by the applicable specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.5 Functional.

4.4.5.1 Initial production sample.-

4.4.5.1.1 Excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit(F). to plus 75 Degrees F. and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.2 Cold temperature phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 55,600 pounds per square inch (PSI) to 57,600 pounds PSI, piezo equivalent pressure, when temperature conditioned at 70 degrees F. The complete round will be temperature conditioned at minus 35 degrees F. to minus 45 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.3 Excess, excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F, to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.2. Production lot sample.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C - Cartridge cases shall be packed in accordance with requirements of Drawing F 7548235.

5.2 Labeling and marking.- Labeling and marking for shipment shall be in accordance with MIL-P-10025.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

5.3 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Prior to fabrication of an initial production sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.3 Ejection.- The term"ejection" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

6.4 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

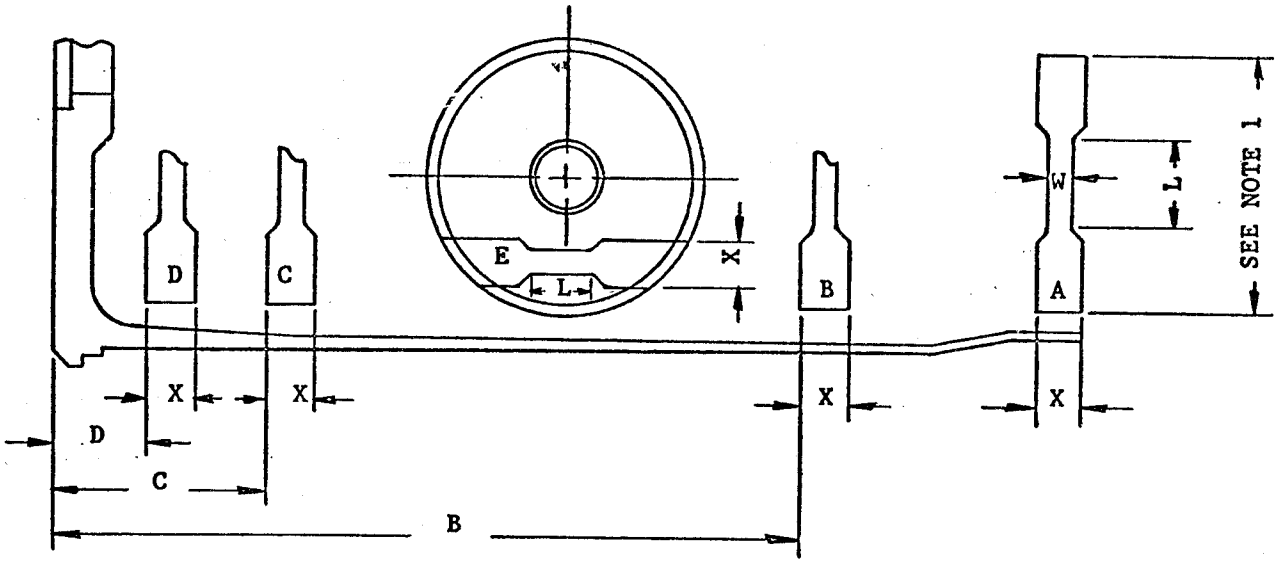
Army - MU

Preparing activity:

Army - MU

Project no. 1315-A499

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.



SPECIMEN	X	W	L	LOCATION FROM HEAD
A	.40	.25	1.0	At Mouth
B	.40	.25	1.0	15.25
C	.40	.25	1.0	2.00
D	.40	.25	1.0	0.75
E	.40	.25	1.0	In Head

NOTES:

1 - Overall length of specimen to suit fixture in testing machine.

2 - Specimens C and D to be machined to uniform thickness parallel to inside surface.

3 - Specimen E to be taken from center of head midway between primer hole counterbore and edge of flange.

4 - Specimen E to be machined so as to eliminate effects of stamping on test.

5 - Yield strength in 0.2 percent offset.

6 - Percent elongation in 1.0 inch gage length

STANDARD POSITIONS AND SIZES OF
TENSILE SPECIMENS FOR CASE, CARTRIDGE, 90MM, M71

FIGURE I

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.